



**HỌC VIỆN
NGÂN HÀNG**
BANKING ACADEMY OF VIETNAM

Web Design

Lecture 13 - Web Development Tools

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8.1. Overview of web deployment

8.2. Git & Github

8.3. Domain & Hosting

8.4. Deployment & Testing

8.1. Overview of web deployment



Deploying a website is the process of transferring website source code from the development environment (**local**) to the production environment (**server**) so that users can access it via the Internet.

Objectives:

- Ensure the website **operates stably**.
- Make the website **easy to update** and maintain.



Prepare the source code
files, test them locally.

→ Finalize HTML, CSS, and JS

Manage the source code
changes in the code.

→ Use Git to store and track

Choose a hosting platform
VPS, or a private server.

→ Examples: Vercel, Netlify,

Connect a domain (if available)
for easier user access.

→ Configure the domain name

Upload or sync the source code to the server → Using Git, FTP, or the hosting platform's built-in deployment tools.

Test the website after deployment
devices and fix any issues found.

→ Test on multiple



Git intro

Git concepts

Github

Github basics

Git GitHub Pages

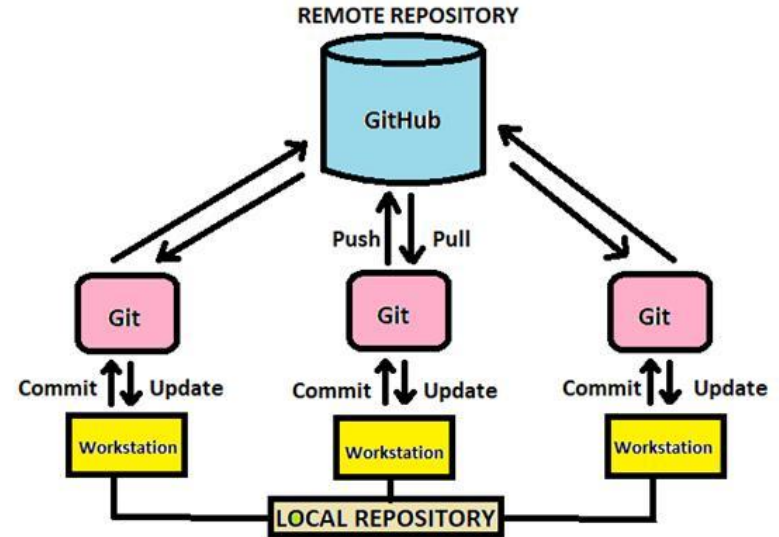


Git is a **version control system**.

Git helps you **keep track of code changes**.

Git is used to **collaborate on code**.

Git and GitHub are different things.





- Manage projects with **Repositories**
- **Clone** a project to work on a local copy
- Control and track changes with **Staging** and **Committing**
- **Branch** and **Merge** to allow for work on different parts and versions of a project
- **Pull** the latest version of the project to a local copy
- **Push** local updates to the main project



- Over **70% of developers** use Git!
- Developers can **work together** from anywhere in the world.
- Developers can **see the full history** of the project.
- Developers can **revert to earlier versions** of a project.

https://www.youtube.com/watch?v=e9lnsKot_SQ



Download Git for free from the following website:

<https://www.git-scm.com/>

- For Windows, you can use **Git bash**, which comes included in **Git for Windows**.
- For Mac and Linux you can use the **built-in terminal**.

Configure Git

- `git config --global user.name "w3schools-test"`
- `git config --global user.email "test@w3schools.com"`



Install Git for Windows on your machine

Register a Github account (if needed)

Configure Git global username & email

- Duration: 10 minutes



- Initialize Git
https://www.w3schools.com/git/git_getstarted.asp?remote=github
- Adding New Files
https://www.w3schools.com/git/git_new_files.asp?remote=github
- Staging Environment
https://www.w3schools.com/git/git_staging_environment.asp?remote=github
- Commit
https://www.w3schools.com/git/git_commit.asp?remote=github
- Branch
https://www.w3schools.com/git/git_branch.asp?remote=github
- Merge Branches
https://www.w3schools.com/git/git_branch_merge.asp?remote=github

Video tutorial:

https://www.youtube.com/watch?v=MFtsLRphqDM&list=PL4cUxeGkcC9goXbgTDQ0n_4TBzOO0ocPR&index=2



https://www.w3schools.com/git/git_quiz.asp?remote=github

What is GitHub?



- Git is not the same as GitHub.
- GitHub makes tools that use Git.
- GitHub is the largest host of source code in the world, and has been owned by Microsoft since 2018.



Register your Github account

Create a directory for your group project

Push Local Repository to GitHub



Pulling to Keep up-to-date with Changes

- Git Fetch
- Git Merge
- Git Pull

https://www.w3schools.com/git/git_pull_from_remote.asp?remote=github



Push Changes to GitHub

https://www.w3schools.com/git/git_push_to_remote.asp?remote=github



Create a New Branch on GitHub

https://www.w3schools.com/git/git_remote_branch.asp?remote=github

Pulling a Branch from GitHub

https://www.w3schools.com/git/git_branch_pull_from_remote.asp?remote=github

Push a Branch to GitHub

https://www.w3schools.com/git/git_branch_push_to_remote.asp?remote=github



With GitHub pages, GitHub allows you to host a webpage from your repository.

1. Create a New Repository
2. Push Local Repository to GitHub Pages
3. Check Out Your Own GitHub Page

https://www.w3schools.com/git/git_remote_pages.asp?remote=github

https://www.w3schools.com/git/git_exercises.asp?remote=github

<https://docs.github.com/en/pages/quickstart>



https://www.w3schools.com/git/git_exercises.asp?remote=github

Push your local project to a Github repo, then create a Github page from the repo

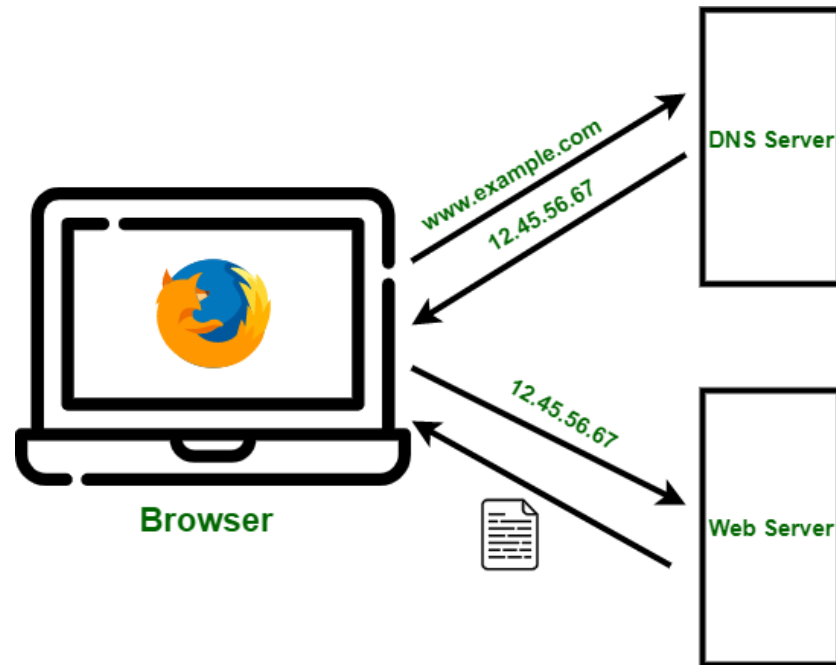
- Duration: 20 mins
- 3 nhóm đầu tiên ở mỗi lớp trình bày: cộng 1 contribution
- Các nhóm không trình bày: -1 contribution
- Mỗi nhóm max 2 phút
 - Show nhanh mã nguồn trang web hiện tại của nhóm trên máy local (VS Code mở cả project folder)
 - Show repo của nhóm trên Github, những lần commit, mỗi thành viên đã commit những gì
 - Show trang web đã up lên Github page



How DNS works

Domain name best practices

Choose a hosting





1. The user enters a web address or domain name into a browser.
2. The browser sends a message, called a **recursive DNS query**, to the network to find out which IP or network address the domain corresponds to.
3. The query goes to a **recursive DNS server**, which is also called a recursive resolver, and is usually managed by the **internet service provider (ISP)**. If the recursive resolver has the address, it will return the address to the user, and the webpage will load.
4. If the recursive DNS server does not have an answer, it will query a series of other servers in the following order: **DNS root name servers, top-level domain (TLD) name servers** and **authoritative name servers**.
5. The three server types work together and continue redirecting until they retrieve a DNS record that contains the queried IP address. It sends this information to the recursive DNS server, and the webpage the user is looking for loads.
6. The recursive server **stores**, or **caches**, the **A record** for the domain name, which contains the IP address. The next time it receives a request for that domain name, it can respond directly to the user instead of querying other servers.
7. If the query reaches the authoritative server and it cannot find the information, it returns an **error message**.



1. Stay Unique And On Brand
2. Choose Your Top-Level Domain Extension Carefully
3. Incorporate Targeted Keywords
4. Make Sure Your Domain Is Easy To Pronounce And Spell
5. Avoid Hyphens, Abbreviations, And Doubled Letters

6. Keep Your Domain Name Short
7. Pick A Domain Name That's Flexible
8. Check The Domain History
9. Research Social Media Handles
10. Protect Your Brand With Multiple Domains

<https://www.dreamhost.com/blog/how-to-choose-the-right-domain-name/>



What is your topic for the final project website?
Think of 3-5 domain names that could be used
Check the availability of those domain names



<https://www.hostinger.com/tutorials/how-to-choose-a-web-hosting-provider>

Not too important at this moment, but required for the Web Programming course

Any alternative? List some of them and get contribution points



Deploy your site on Vercel

Deploy to other hosting

- FileZilla

- WinSCP

Test the visualization on multiple devices and make some modifications if needed

Register with search engines

- <https://www.google.com/webmasters/tools>

- <https://www.bing.com/webmasters/about>



<https://www.youtube.com/watch?v=X6boluWzBhg>